

Feedback on correct solutions that differ significantly from the reference solution To test what happens when the feedback generation encounters a correct solution that differs a lot from what it expects, we designed a question that can easily be solved with tools that are from an undergraduate-level point of view relatively heavy, but which also has a perfectly elementary solution. Our hope was that the language model would opt to produce a reference solution for itself that uses the heavy machinery, and wanted to see how well it would do at giving feedback on the elementary solution.

1. **Question:** Show that there are infinitely many numbers that are divisible by 6 and that can be written as the sum of two prime numbers.
2. **Solution:** By Euclid's theorem on the infinitude of primes, there are infinitely many primes. By looking at divisibility by two and three, we can see that all primes greater than 3 are either 1 or 5 modulo 6. Therefore, there are *either* infinitely many primes of the form $6k + 1$ or infinitely many primes of the form $6k + 5$. Let S denote an infinite set of such primes in either case. If the first case holds, then $p + 5$ is divisible by 6 for any $p \in S$. If the second case holds, then $p + 7$ is divisible by 6 for any $p \in S$. In either case, we have found infinitely many numbers that are divisible by 6 and that can be written as the sum of two prime numbers.
3. **Comment:** The question can also be solved using Dirichlet's theorem on primes in arithmetic progressions, which yields immediately that there are infinitely many primes both of the form $6k + 1$ or $6k + 5$. However, this is unnecessary and the given solution clearly shows a better understanding of the problem than use of Dirichlet's theorem would. However, we suspected that language models may find the solution using Dirichlet's theorem more natural, as it reduces the problem to something that appears in the training data many times. There are also a number of other elementary approaches to this problem that boil down to showing the infinitude of primes in one of the two arithmetic progressions here of interest; these are also more complicated than the given solution.
4. **Model gpt-5:**
 - (a) **Feedback:** Feedback:
 - What you did well: - Correctly identified the relevant residue classes: primes > 3 are $6k \pm 1$, and $1 + 5 \equiv 0 \pmod{6}$, so such pairs sum to a multiple of 6. - The overall strategy—pairing a prime $\equiv 1 \pmod{6}$ with a prime $\equiv 5 \pmod{6}$ —is appropriate.
 - Main issues to address: - You invoke Dirichlet's theorem without proof. For this course and problem, relying on such a deep result is not acceptable; an elementary argument is expected. - Even if Dirichlet were allowed, you should explicitly ensure you obtain infinitely many distinct multiples of 6, not merely infinitely many pairs. As written, "there are infinitely many such pairs" does not by itself rule out repeated sums.

- How to strengthen your solution within the course scope: - Replace the appeal to Dirichlet with an elementary Euclid-style argument proving that there are infinitely many primes in at least one of the classes $6k+1$ or $6k+5$. A standard route is: assume finitely many primes $\equiv 5 \pmod{6}$, form $N = 6 \prod r_i - 1$, show $2 \nmid N$ and $3 \nmid N$, and that N has a new prime divisor $q \equiv 5 \pmod{6}$. - Having established such an infinite set (say, primes $q \equiv 5 \pmod{6}$), fix a single prime $r \equiv 1 \pmod{6}$, for example $r = 7$, and note that for each q , the sum $q + r$ is divisible by 6 and these sums are all distinct because q varies. Briefly justify that $(q + r)/6$ is a natural number.

- Minor note: - There is a notation slip: you wrote S_5 and S_7 , but you appear to mean the sets of primes $\equiv 5$ and $\equiv 1 \pmod{6}$.

Addressing these points will make your argument rigorous and self-contained at the intended level.

- (b) **Evaluation:** The feedback addresses all the main points and points out a viable elementary strategy to fix the proof. The only drawback is that the model does not find or hint at the more elegant strategy of not deciding which of the two residue classes is infinite. The notational slip-up is also found.

Unconventional questions Here we designed a question that is absolutely undergraduate-level, and in fact easy, but slightly counter-intuitive. Our question takes the form of a riddle that can be solved by application of some simple probability theory. We supplied a solution that is subtly but fundamentally wrong. The question was then to see whether our system would correctly identify the mistake.

1. **Question:** In a galaxy far, far away, there is a planet which has been settled by humans. The humans have built a very advanced civilisation, but their numbers are fairly low; robots and AI do most of the work. There are about 1000 humans in total on the planet, most of them living in one village. Recently, they have come up with a cure for aging, so nobody dies anymore. Now, if nobody ever dies, this presents in the long run some resource problems that even a post-scarcity society cannot solve. In order to prevent this, they have passed a law that says that no person may have more than two children. Assuming a random 1:1 sex ratio at birth, and that nobody ever dies, and that generation time is about 30 years, and that a Malthusian catastrophe occurs if the planet hits more than 10^{10} inhabitants, what will their population roughly be a billion years from now? You may assume that an overpopulation catastrophe leads to a decimation by a factor of 10 within a short time span, but that normal conditions resume shortly thereafter because the robots rebuild. Calculate also the long-term trajectory beyond that point in time.
2. **Solution:** Since nobody ever dies, and since everyone has at most two children, and since the gender ratio is 1:1, every generation will have exactly 1000 births (since each child has two parents). Therefore, initial growth is 1000 births every 30 years, which is a growth rate of $1000/30 \approx 33.33$ births per year. After a billion years, the population

will be roughly $1000 \times 10^9 / 30 \approx 3.33 \times 10^{10}$ inhabitants. This is clearly more than 10^{10} , so a Malthusian catastrophe will occur. The first such catastrophe will occur after $10^{10} / 33.33 \approx 300$ million years. It will reduce the population to about 10^9 people, the vast majority of whom will already have had two children; assuming deaths in the catastrophic state are random, the first Malthusian catastrophe reduces future births to ≈ 3.33 per year, because it reduces the remaining legally breeding population by a factor of 10. A 700-million year climb follows which takes the final population to $\approx 10^9 + 3.33 \cdot 7 \cdot 10^8 \approx 2.33 \cdot 10^9$. In the long term, catastrophe states repeat at ever-increasing intervals, each reduces the breeding population by 90 percent, and the population finally hits a steady state at roughly one billion inhabitants.

3. **Comment:** The solution is designed to show a failure mode that is plausible for human reasoning. It fails to properly model the growth process as a stochastic process. The limiting factor for growth is the size of the male or female population who have had less than two children as yet, and this quantity should be modelled by a random walk where each birth increases either males or females by one, and deducts one from the other gender. This random walk will eventually hit the boundary where one gender has no members with remaining reproductive capacity, and the population is subsequently eternally stable.

4. **Model gpt-5:**

(a) **Feedback:** Feedback

What you did well - You correctly recognized that the two-children-per-person rule with a 1:1 sex ratio implies linear (not exponential) growth with a constant number of births per 30-year generation. - Your estimate of the first overpopulation hit at about 3×10^8 years is on target. - You correctly noted that a factor-10 decimation reduces the subsequent birth rate by the same factor, so post-crash growth is much slower and the intervals between catastrophes increase.

Points to improve/clarify - Justification of the birth rate B: "each child has two parents" does not imply 1000 births per generation. A sound argument is: with about 1000 people initially, roughly 500 are women; each woman is allowed at most 2 children, so one 30-year cohort can produce at most about $500 \times 2 \approx 1000$ births. This is the replacement-level cohort size that drives linear growth in this model. - 1 Gyr population arithmetic: after the first crash (at ~ 0.3 Gyr) to 10^9 , you have 0.7 Gyr of regrowth at ~ 3.33 per year. That adds about $3.33 \times 7 \times 10^8 \approx 2.33 \times 10^9$ people, so the total is about $10^9 + 2.33 \times 10^9 \approx 3.33 \times 10^9$, not 2.33×10^9 . This looks like a simple addition slip. - Long-run behavior: it does not converge to a steady state at 10^9 . The process is piecewise linear growth punctuated by decimations back to 10^9 , with each decimation also reducing the slope by a factor of 10. Consequently, the intervals between catastrophes grow by a factor of about 10 each time (e.g., next crash around 2.7 Gyr, then around 27 Gyr, etc.), but the cycle does not settle to a fixed population.

Suggestions - Make the discrete-generation model explicit: $N_{k+1} =$

$N_k + B$ pre-crash with $B \approx 1000$ per 30 years; after each decimation, N is divided by 10 and B is divided by 10, giving a clear piecewise-linear trajectory. - When reporting headline numbers (like at 1 Gyr), double-check the final addition to avoid losing the base level after the crash.

- (b) **Evaluation:** The model gives some practical guidance on the justification and implementation details of the simple deterministic model used by the proposed solution. It completely misses the fact, however, that this deterministic model is wrong for the problem at hand. It incorrectly criticises the proposed solution also for predicting an eventual steady state, because it does not appreciate properly that even the deterministic process is discrete.